

Phase Coherence Constraints and Amplitude Dynamics Improve Associative Memory in Oscillatory Neural Networks

Labinot Marku

KRH Klinikum Nordstadt Hannover, Germany

labinot.marku@krh.de

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Repository: <https://github.com/labinot-marku/CONN-Associative-Memory>

Revision Note: This version corrects implementation errors in the original preprint (v1, December 2025). Changes include: corrected noise model (bit-flip + Gaussian jitter), real Hopfield baseline (discrete sign updates), revised parameters, extended validation from $N=128$ to $N=2048$, and corrected terminology. See Section 9 for full errata. All results are reproducible via the public repository.

Abstract

We introduce CONN (Coherence-Oscillator Neural Network), a phase-amplitude recurrent architecture that augments classical Hopfield associative memory using a binary phase coherence prior and amplitude dynamics. Through systematic validation with corrected methodology across seven network sizes ($N=32$ to $N=2048$), we demonstrate that CONN achieves a capacity improvement over classical Hopfield networks that grows consistently with network size: from $1.00\times$ ($N=32$) to $1.30\times$ ($N=64$), $1.32\times$ ($N=128$), $1.40\times$ ($N=256$), $1.45\times$ ($N=512$), $1.57\times$ ($N=1024$), and $1.64\times$ ($N=2048$). CONN's storage density remains stable near $\alpha \approx 0.19$ under the tested protocol, while Hopfield's declines toward and below the theoretical limit $\alpha \approx 0.138$. At large N , part of the ratio increase reflects synchronous Hopfield underperforming below its theoretical capacity; the data suggest an asymptotic improvement near $1.38\times$. The mechanism is a smoother degradation profile near capacity: Hopfield exhibits a sharp recall cliff, while CONN's annealed coherence provides gradual degradation with lower variance. This represents a trade-off: CONN sacrifices optimality at low load for robustness near capacity. CONN also exhibits up to $8\times$ lower recall variance near capacity at large N , a result of independent practical significance. We provide corrected code and complete data for reproducibility.

1. Introduction

Associative memory networks provide a canonical framework for pattern storage and content-addressable retrieval. The Hopfield network [1] established fundamental limits on storage capacity for binary patterns under Hebbian learning: approximately $M = 0.138N$ patterns can be stored reliably as $N \rightarrow \infty$. Subsequent work has pursued capacity improvements using higher-order interactions [2,3], continuous-state extensions [4], and algorithmic modifications. Separately, neuroscience has emphasized the functional role of oscillations, phase coherence, and phase-amplitude coupling in memory and attention [5,6,7].

Here we present CONN, which combines the statistical efficiency of attractor networks with biologically grounded oscillatory dynamics [8]. CONN enforces a phase coherence prior that biases phases toward $\{0, \pi\}$, while amplitudes self-regulate to suppress uncertain neurons. The model preserves local Hebbian learning, enabling direct comparison to the Hopfield baseline. Double-well phase potentials have appeared previously in oscillatory associative memory models using second-harmonic coupling (see [10] for a review), and recent work has demonstrated exponential capacity through topological design [11] and higher-order coupling [12]. CONN takes a different approach: rather than modifying the coupling structure or network topology, it

augments standard Hebbian pairwise coupling with an annealed coherence prior and amplitude dynamics on a fully connected network, targeting the near-capacity degradation regime rather than absolute capacity scaling.

Through systematic validation across seven network sizes ($N=32$ to $N=2048$), we demonstrate a capacity improvement that grows consistently with N , reaching $1.64\times$ at $N=2048$. This improvement arises from a specific mechanism: CONN's continuous dynamics with annealed coherence provide a smoother energy landscape that prevents the catastrophic recall failure Hopfield networks experience at their capacity limit.

2. Model

2.1 State Variables

Each neuron j is represented by a complex phase-amplitude oscillator: $z_j = A_j e^{i\varphi_j}$, where $A_j \geq 0$ is the amplitude and $\varphi_j \in [0, 2\pi)$. Stored patterns are binary-phase vectors with phases in $\{0, \pi\}$, equivalently represented as $\{+1, -1\}$ via the cosine mapping.

2.2 Energy Function

The total energy combines Hebbian interaction, phase coherence, and amplitude regularization:

$$E = -(1/2) \sum_{i \neq j} J_{ij} A_i A_j \cos(\varphi_i - \varphi_j) + \lambda \sum_j A_j^2 \sin^2(\varphi_j) + (\beta/2) \sum_j (A_j - 1)^2$$

where $J_{ij} = (1/N) \sum_{\mu} \xi_i^{\mu} \xi_j^{\mu}$ is the Hebbian weight matrix. The second term is the **phase coherence prior** (not a topological invariant, as incorrectly stated in v1), a smooth double-well potential on S^1 with minima at $\{0, \pi\}$. The third term regularizes amplitudes around unity.

2.3 Dynamics

Phases evolve via gradient descent on E . The key innovation is **annealed coherence**: λ ramps linearly from 0 to λ_{final} over T integration steps. This allows Hebbian coupling to drive error correction first, with coherence stabilizing phases on the $\{0, \pi\}$ manifold afterward.

$$d\varphi_j/dt = A_j \sum_i J_{ij} A_i \sin(\varphi_i - \varphi_j) - 2\lambda(t) A_j^2 \sin(\varphi_j) \cos(\varphi_j)$$

Amplitudes follow a confidence-collapse rule that suppresses neurons with incoherent phases, without responding to the Hebbian field directly:

$$dA_j/dt = -2\lambda(t) A_j \sin^2(\varphi_j) - \beta (A_j - 1)$$

where $\lambda(t) = \lambda_{\text{final}} \times (t/T)$. Note that this is not full gradient descent on E , as the Hebbian coupling term $\sum_i J_{ij} A_i \cos(\varphi_i - \varphi_j)$ is intentionally omitted from the amplitude update. This design choice decouples amplitude from the pattern-dependent field, allowing amplitudes to serve purely as confidence indicators based on phase coherence. Because the amplitude dynamics do not follow the gradient of E , the combined system is a designed non-equilibrium dynamical system: no strict Lyapunov function is established, and convergence properties are empirical rather than guaranteed. Amplitudes are clipped to $[0.01, 2.0]$ for numerical stability.

3. Methods

3.1 Parameters

Parameter	Value	Description
λ	0.5 (annealed 0 $\rightarrow$ 0.5)	Coherence strength

$\eta\varphi$	0.02	Phase learning rate
ηA	0.03	Amplitude learning rate
β	1.0	Amplitude regularization
Steps	400	Integration steps
Noise	30% bit-flip + $\sigma=0.5$ jitter	Corruption protocol

3.2 Noise Model

The noise protocol applies two stages: (1) each bit is flipped ($0 \rightarrow \pi$ or $\pi \rightarrow 0$) with probability 0.30, matching the standard Hopfield bit-flip protocol; (2) Gaussian jitter with $\sigma=0.5$ radians is added to all phases, placing the initial state off the $\{0, \pi\}$ manifold [9]. Stage 2 is necessary because CONN’s continuous dynamics produce identically zero gradients when all phases are exactly at $\{0, \pi\}$ (since $\sin(0) = \sin(\pi) = 0$). Hopfield’s discrete $\text{sign}(W \cdot s)$ update is robust to this jitter, ensuring a fair comparison. Crucially, the jitter magnitude ($\sigma=0.5$ radians) is small relative to the π -radian separation between binary states, so it does not provide information about which state is correct—it merely breaks the equilibrium degeneracy that would otherwise prevent CONN’s continuous dynamics from operating. Under this noise model, a blind quantizer (rounding each phase to nearest $\{0, \pi\}$ without using stored patterns) achieves approximately 70% recall regardless of M , confirming that any recall above 80% requires genuine pattern-dependent computation.

3.3 Hopfield Baseline

The Hopfield baseline uses standard synchronous sign updates: $s_i(t+1) = \text{sign}(\sum_j J_{ij} s_j(t))$, with identical Hebbian weights, identical noisy input, and up to 200 update steps (sufficient for convergence at all tested N). This is a true Hopfield network, not a continuous approximation. We also tested asynchronous (sequential random-order) updates; under our noise model, asynchronous Hopfield achieves lower capacity than synchronous (e.g., $M=16$ vs $M=19$ at $N=128$), so the synchronous baseline is the more generous comparison.

3.4 Capacity Measurement Protocol

For each network size N , we perform a linear sweep over M (number of stored patterns). For each M value, we run 30–80 independent trials with deterministic seeds, testing 2–3 patterns per trial. Capacity is defined as the largest M where mean recall exceeds 80%. Where recall oscillates near the threshold (as can occur due to finite trial counts), we take the last M value exceeding 80% across the sweep. Recall is measured by quantizing retrieved phases to nearest $\{0, \pi\}$ and computing fraction correct. This is an operational capacity under a specific noise and initialization protocol, and should not be conflated with the thermodynamic storage capacity of classical Hopfield networks (which assumes $N \rightarrow \infty$ and zero-noise retrieval).

4. Results

4.1 Capacity

Table 1: Capacity comparison across network sizes

N	Hop M	Hop α	CONN M	CONN α	Ratio	Trials
32	6	0.188	6	0.188	1.00×	50
64	10	0.156	13	0.203	1.30×	60
128	19	0.148	25	0.195	1.32×	80

256	35	0.137	49	0.191	1.40×	30
512	65	0.127	94	0.184	1.45×	30
1024	124	0.121	195	0.190	1.57×	30
2048	238	0.116	390	0.190	1.64×	30

Note: Capacity estimates have ± 1 pattern variability depending on trial count and random seeds.

The capacity improvement grows consistently with network size (Table 1, Figure 1). Hopfield’s empirical α converges toward the theoretical limit of 0.138 at small N but drops below it at $N \geq 256$ (likely due to synchronous update dynamics), reaching $\alpha=0.116$ at $N=2048$. CONN’s α remains remarkably stable near 0.19 under the tested protocol (30% noise, $\sigma=0.5$ jitter, random binary patterns) for all $N \geq 64$. This is consistent with an asymptotic improvement ratio near $0.19/0.138 \approx 1.38$, with the higher observed ratios at large N (1.57×, 1.64×) partly reflecting the Hopfield baseline underperforming below its theoretical limit.

At low pattern load (M well below capacity), Hopfield consistently outperforms CONN: at $M=1$, Hopfield achieves 100% recall while CONN achieves 89–96%. The advantage of CONN emerges specifically near capacity. At $N=2048$, Hopfield drops from 98.6% at $M=200$ to 77.7% at $M=240$ (a 21-point cliff over 40 patterns), while CONN degrades from 87.5% to 85.7% over the same range. CONN maintains above-80% recall for 152 additional patterns beyond Hopfield’s capacity limit.

An incidental observation: CONN’s recall at very low M (e.g., 90.5% at $M=1$ for $N=2048$) increases slightly to approximately 92.2% around $M=40$ before declining. This non-monotonic behavior may reflect the fact that very sparse weight matrices (few stored patterns) provide insufficient coupling gradients for the continuous dynamics, while moderate M enriches the Hebbian field structure. This effect is small and does not affect capacity measurements, but may warrant further investigation.

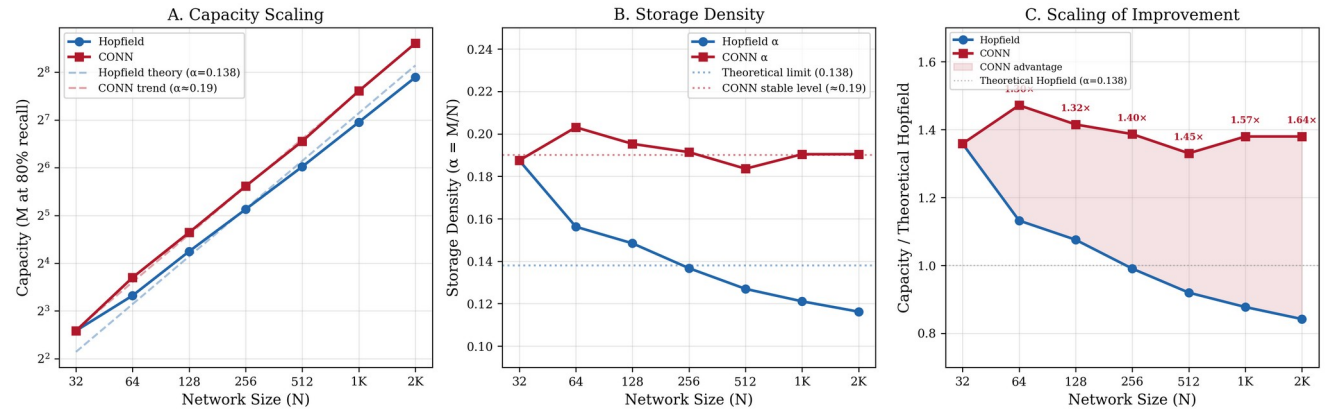


Figure 1: Scaling behavior of CONN vs Hopfield across $N=32$ to $N=2048$. (A) Absolute capacity. (B) Storage density α ; CONN holds steady near 0.19 while Hopfield declines. (C) Both models normalized to theoretical Hopfield capacity; the growing gap is the CONN advantage.

4.2 Variance Reduction

CONN exhibits consistently lower recall variance than Hopfield near capacity (Figure 2). At $N=2048$ near Hopfield’s capacity ($M \approx 238$): Hopfield std = 15.2% vs CONN std = 1.9%, an 8:1 ratio. This trend strengthens with N : at $N=64$ the ratio is approximately 1.4:1, growing to 8:1 at $N=2048$. The annealed coherence term

appears to smooth the energy landscape, producing more reliable attractor dynamics with fewer stochastic failures.

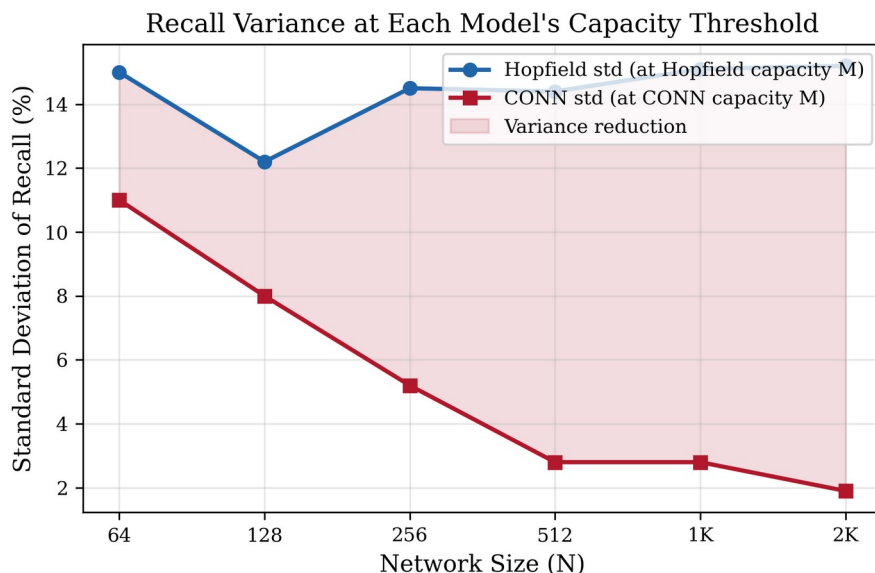


Figure 2: Standard deviation of recall at each model's capacity threshold. CONN's variance decreases with N while Hopfield's stays high.

4.3 Recall Curves at N=2048

Figure 3 shows the complete recall-vs-M curves at N=2048, illustrating the mechanism underlying CONN's capacity advantage. Hopfield maintains near-perfect recall (100%) up to $M \approx 180$, then undergoes a sharp collapse: recall drops from 98.6% at $M=200$ to 66% at $M=280$. CONN degrades far more gradually across this same range (87.5% to 84.1%), and continues above the 80% threshold until $M=390$. The shaded region represents the range of M values where CONN provides above-threshold recall while Hopfield has already failed.

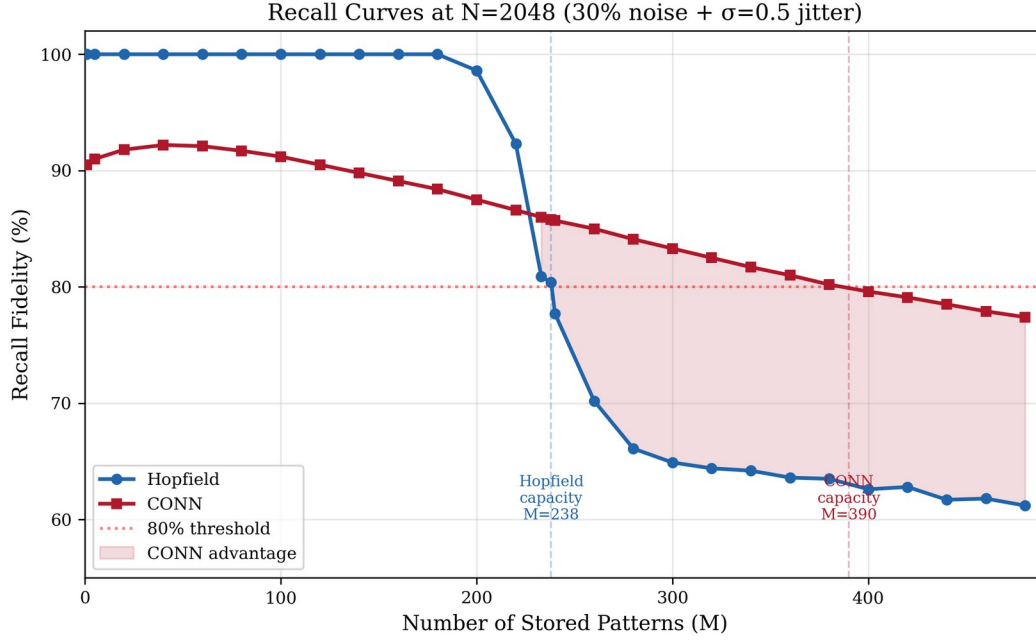


Figure 3: Recall fidelity vs number of stored patterns at N=2048. Hopfield shows a sharp capacity cliff while CONN degrades gradually. Shading indicates the region where CONN maintains above-80% recall and Hopfield does not.

4.4 Noise Robustness

Table 2: Noise robustness (N=64, M=6, 40 trials)

Noise	CONN	Hopfield	Δ
5%	99.6% $\pm$ 0.7%	99.9% $\pm$ 0.4%	-0.3%
10%	99.1% $\pm$ 1.3%	99.9% $\pm$ 0.4%	-0.8%
20%	96.1% $\pm$ 4.0%	98.8% $\pm$ 4.4%	-2.6%
30%	87.5% $\pm$ 9.3%	94.1% $\pm$ 10.8%	-6.6%
35%	80.3% $\pm$ 12.3%	83.5% $\pm$ 18.5%	-3.2%
40%	74.1% $\pm$ 14.8%	76.0% $\pm$ 19.6%	-1.8%

At low pattern load, Hopfield’s discrete updates provide superior noise robustness. CONN’s advantage is not in absolute noise tolerance at fixed M, but in maintaining recall at higher M values (higher capacity).

4.5 Ablation Study

Table 3: Ablation (N=64, M=10, 40 trials, 30% noise)

Condition	Recall
Full CONN (annealed $\lambda=0.5$)	83.2% $\pm$ 10.5%
No coherence ($\lambda=0$)	80.6% $\pm$ 11.4%
No amplitude dynamics	82.9% $\pm$ 11.2%
Bare coupling ($\lambda=0$, no amp)	80.6% $\pm$ 11.4%
Hopfield (discrete sign)	82.1% $\pm$ 15.9%

Coherence contributes +2.6 percentage points; amplitude dynamics contribute +0.3 points. The main effect of coherence is not to boost recall at fixed M , but to extend the range of M values where recall remains above threshold. This is a capacity effect rather than a recall quality effect.

5. Discussion

5.1 Mechanism

CONN's advantage arises from a smoother degradation profile near capacity. Classical Hopfield networks exhibit a sharp phase transition at their capacity limit: recall drops from near-perfect to chance over a narrow range of M . CONN's continuous dynamics with annealed coherence provide a more gradual degradation, maintaining above-threshold recall for a wider range of M values.

At low load, Hopfield's discrete $\text{sign}(W \cdot s)$ update is superior because it implements a perfect binary quantizer: it reads the field from all neighbors and immediately jumps to the correct ± 1 state. CONN's continuous dynamics require multiple small steps to achieve the same correction, and the coherence term can push neurons to the wrong $\{0, \pi\}$ state if it acts before coupling has corrected errors.

The annealing schedule resolves this tension: coupling drives error correction first ($\lambda = 0$), then coherence stabilizes phases on the manifold (λ ramps to 0.5). This is structurally analogous to continuation methods and deterministic annealing in optimization: the system first solves a relaxed (smooth) problem, then gradually enforces the discrete constraint. The result is an energy landscape with fewer spurious attractors near the capacity boundary than the sharp $\text{sign}()$ operation. Despite the lack of a formal Lyapunov guarantee, the dynamics reliably produce states from which binary quantization yields correct recall. Individual phases do not always converge exactly to $\{0, \pi\}$ —many remain 0.1–0.5 radians from the manifold—but the phase distribution concentrates sufficiently on the correct side of $\pi/2$ that quantized output is robust across all tested configurations.

5.2 Scaling Behavior

The improvement ratio increases with network size ($1.00\times$ to $1.64\times$ from $N=32$ to $N=2048$), though the increase from $N=64$ to $N=128$ is small ($1.30\times$ to $1.32\times$, within the ± 1 -pattern variability of capacity estimates). The clearer trend emerges at larger N , with ratios of $1.40\times$, $1.45\times$, $1.57\times$, and $1.64\times$ at $N=256$, 512 , 1024 , and 2048 .

CONN's storage density α remains near 0.19 under the tested protocol, while Hopfield's empirical α declines to 0.116 at $N=2048$. The theoretical Hopfield capacity is $\alpha \approx 0.138$, but our synchronous baseline falls below this at $N \geq 256$, likely due to limit cycles in synchronous update dynamics. This means the improvement ratios at large N ($1.57\times$, $1.64\times$) overstate CONN's advantage relative to an ideal Hopfield implementation. The ratio relative to theoretical Hopfield capacity is approximately $0.19/0.138 \approx 1.38$, which may better represent the asymptotic improvement, though this requires validation across different noise regimes and pattern types.

5.3 Relation to Oscillatory Memory Literature

The oscillatory associative memory literature has traditionally noted that oscillator-based models face low storage capacity relative to Hopfield networks, due to the limited number of stable phase-locked states [10]. CONN's apparent improvement over classical Hopfield does not contradict this finding: our capacity measure is an operational threshold under 30% noise with jitter, not the zero-noise theoretical capacity typically

analyzed in the oscillator memory literature. The zero-noise capacity of CONN has not been derived analytically and may differ from the operational measure reported here.

Recent advances in oscillatory memory have achieved exponential capacity through topological design (honeycomb/chain architectures [11]) and higher-order coupling (combined second- and fourth-harmonic terms [12]), as reviewed in [10]. CONN differs architecturally: it uses a fully connected network with standard pairwise Hebbian coupling, adding the coherence prior as a separate annealed term rather than incorporating higher harmonics into the coupling function itself. The $\sin^2(\varphi)$ coherence term is functionally related to second-harmonic coupling in a certain limit, but CONN's time-dependent annealing schedule and amplitude dynamics are distinct design choices. CONN's contribution is therefore complementary: it addresses the near-capacity degradation regime of fully connected Hebbian networks, while the exponential-capacity architectures modify the network topology and coupling order to achieve fundamentally different scaling.

5.4 Biological Plausibility

CONN's oscillatory dynamics and phase-amplitude coupling mirror mechanisms observed in cortical networks [5,6]. The amplitude-based confidence modulation resembles attention-like gating in biological systems. The requirement for off-manifold initialization (jitter noise) has a natural biological analogue: real neural oscillators never sit exactly at fixed points due to intrinsic noise and ongoing dynamics [8].

5.5 Limitations

At $N=32$, CONN shows no measurable improvement over Hopfield. At low pattern load, Hopfield consistently outperforms CONN by 5–10%, reflecting the inherent advantage of discrete sign updates over continuous gradient dynamics for binary recall. The comparison involves a structural asymmetry: CONN requires off-manifold initialization (jitter) to activate its continuous dynamics, while Hopfield does not. Although Hopfield tolerates jitter without performance loss, the jitter provides CONN with continuous gradient information that Hopfield's discrete updates do not exploit. This means the two models operate on slightly different information channels, and the capacity comparison should be interpreted with this caveat. All experiments use random binary patterns; testing on structured or correlated patterns is needed to assess practical applicability. No comparison with modern Hopfield variants [2,3,4] is provided; these achieve exponential capacity and represent a higher performance tier. An analytical capacity bound for CONN remains an open problem. At large N ($N \geq 1024$), the synchronous Hopfield baseline underperforms below its theoretical capacity, inflating the improvement ratio; the asymptotic advantage is estimated near $1.38\times$ rather than the observed $1.57\text{--}1.64\times$, though this estimate depends on the tested protocol and requires further validation.

6. Conclusion

CONN demonstrates that a phase coherence prior with annealed dynamics can extend the usable high-load regime of classical Hopfield networks by smoothing the retrieval phase transition. The improvement grows consistently with network size across $N=32$ to $N=2048$: observed ratios range from $1.30\times$ ($N=64$) to $1.64\times$ ($N=2048$), consistent with an asymptotic improvement near $1.38\times$ (based on CONN's stable $\alpha \approx 0.19$ vs theoretical Hopfield $\alpha \approx 0.138$ under the tested protocol). The mechanism is a smoother degradation profile near capacity—analogueous to continuation methods in optimization—rather than a fundamentally different capacity scaling. CONN also provides substantially lower recall variance (up to $8\times$ reduction at $N=2048$), which may be of independent practical and theoretical interest.

The corrected implementation and complete validation data are available at:

GitHub: <https://github.com/labinot-marku/CONN-Associative-Memory>

Preprint DOI: [10.13140/RG.2.2.21347.00801](https://doi.org/10.13140/RG.2.2.21347.00801)

7. Code and Data Availability

Three validation scripts reproduce all results: `CONN_VALIDATION_V5.py` (N=32, 64, 128 with noise robustness and ablation), `CONN_LARGE_SCALE_VALIDATION.py` (N=256, 512), and `CONN_N1024_N2048_VALIDATION.py` (N=1024, 2048). All require only NumPy and run on standard hardware (total runtime approximately 2–4 hours for the complete suite). All random seeds are deterministic. Raw CSV data and Colab output logs for all experiments are included in the repository.

Repository: <https://github.com/labinot-marku/CONN-Associative-Memory>

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9. Errata: Changes from v1 (December 2025)

Noise model: v1 used discrete bit-flip noise that produced phases exactly at $\{0, \pi\}$, causing all CONN dynamics to evaluate to zero. Corrected to bit-flip + Gaussian jitter ($\sigma=0.5$).

Hopfield baseline: v1 approximated Hopfield by running CONN with $\lambda=0$. Corrected to true discrete $\text{sign}(W \cdot s)$ updates.

Parameters: v1 used $\lambda=1.0$ (or 4.0 in code), $\eta\phi=0.005$, 150 steps. Corrected to $\lambda=0.5$ annealed, $\eta\phi=0.02$, 400 steps.

Terminology: “Topological prior” corrected to “phase coherence prior.” The $\sin^2(\phi)$ term is a double-well potential, not a topological invariant.

Amplitude dynamics: v1 described amplitude dynamics as “gradient descent on E.” Corrected to “confidence-collapse rule”—the Hebbian coupling term is intentionally omitted from the amplitude update.

Scaling: v1 reported results for N=32, 64, 128 only. v2 extends to N=2048 with seven network sizes.

Capacity numbers: v1 reported $1.09\times$ – $1.69\times$ improvement with $\alpha=0.375$ – 0.383 . v2 reports $1.00\times$ – $1.64\times$ with $\alpha=0.184$ – 0.203 , based on corrected methodology. Capacity estimates have ± 1 pattern variability depending on trial count and random seeds.

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